Lab I

Installation of C++ and Running First Program

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Section B

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Lab Objectives

The objective of this lab was to become familiar with programming tools and version control. In this lab we installed a C++ development environment, created and executed basic programs, and learned how to use GitHub to upload our lab work. We also created a tutorial video explaining the steps.

Software's Used

1. **Dev C++ / VS Code** – for writing and running C++ programs.
2. **GitHub** – for storing and sharing the lab repository online.
3. 3. MinGW-w64 (Recommended Updated Version)

   MinGW-w64 https://www.mingw-w64.org/

   https://www.mingw-w64.org/downloads/

4. . C/C++ Extension for VS Code

1: Installing Dev C++ / VS Code and Running Basic Programs

First, we installed **Dev C++ / Visual Studio Code** on the computer.

Steps:

1. Downloaded the software from the official website.
2. Installed it on the computer.
3. Opened the IDE (Integrated Development Environment).
4. Created a new C++ file.
5. Wrote a simple C++ program.

2: Creating a GitHub Repository

We created a GitHub account and uploaded the lab files.

Steps:

1. Opened **github.com**.
2. Created a new account.
3. Clicked **New Repository**.
4. Named the repository **Programming-Lab**.
5. Uploaded the **Lab-01 files**.
6. Saved the repository online.

GitHub helps in storing code and sharing it with others.

In this lab we learned how to install a programming environment, write and execute a basic C++ program, and upload our work to GitHub. We also practiced explaining the process by creating a tutorial video. This lab helped us understand the basic tools required for programming.

Lab Task 1:

```
#include <iostream>                    // tells the compiler to include input
                                          output stream liberary
using namespace std;                   // allows us to use standard
                                          namespace cout and c in WIthout have
                                          the need to write std:: for each
int main() {                            // main function from where the execution
                                          begins

cout <<"This is Hikmat Ullah Jan!\n";   // output to the display

return 0;                               // tells that program ended successfully

}
```